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REVIEW

Update on Medical Treatments for Essential Tremor: An International Parkinson and Movement Disorder Society Evidence-Based Medicine Review

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ABSTRACT: Background: The first International Parkinson and Movement Disorder Society Evidence-Based Medicine (MDS-EBM) review for essential tremor (ET) was published in 2019; since then, the modified Grading of Recommendations, Assessment, Development, and Evaluation (GRADE) methodology was adopted by MDS, and new evidence exists.

Objective: The objective of this study was to update EBM conclusions for medical treatments of ET with a focus on longer follow-up periods.

Methods: A systematic literature search was conducted for randomized controlled trials (RCTs) investigating medical interventions for ET with a minimum 1-month follow-up, with subsequent appraisal of data using an MDS-EBM framework.

Results: Thirty-one RCTs were included evaluating 16 interventions against placebo. Nine interventions were evaluated by more than one RCT: alprazolam, botulinum toxin type A (BtA), levetiracetam, phenobarbitone, pregabalin, primidone, propranolol, topiramate, and trazodone. The remainder were studied in a single RCT (acetazolamide, flunarizine, gabapentin, mirtazapine, perampanel, progabide, and zonisamide). Trial sample size ranged from 5 to 117 participants, and study duration ranged from 4 to 28 weeks. More than one RCT documented improvement in tremor severity for propranolol, primidone, topiramate, and BtA. Using the modified GRADE framework, we found significant methodological shortcomings in the studies, resulting in insufficient evidence for all interventions. Concerns about risk of bias and imprecision commonly limited the ability to make stronger recommendations for these interventions.

Conclusions: Current evidence from RCTs with at least 1 month of follow-up is insufficient to confidently support the efficacy of available medical treatments for ET. There is a need for longer, higher-quality clinical trials to improve treatment recommendations and guide decision-making for clinicians and patients with ET. © 2026 The Author(s). *Movement Disorders* published by Wiley Periodicals LLC on behalf of International Parkinson and Movement Disorder Society. © 2026 The Author(s). *Movement Disorders* published by Wiley Periodicals LLC on behalf of International Parkinson and Movement Disorder Society.

Key Words: essential tremor; evidence-based medicine; GRADE; medical treatment

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Financial disclosures and conflicts of interest: Author disclosures are available in the Supporting Information.

Received: 9 October 2025; **Revised:** 2 December 2025; **Accepted:** 18 December 2025

Published online in Wiley Online Library (wileyonlinelibrary.com). DOI: 10.1002/mds.70184

Essential tremor (ET) is defined as a bilateral, symmetric action tremor of the hands/forearms lasting at least 3 years, without other neurological signs.[1,2] ET is one of the most common movement disorders with an estimated prevalence of 0.32%–0.9% worldwide.[3] ET can have a significant impact on daily functioning, hindering independence in both personal and professional life, and can be associated with psychosocial effects such as increased anxiety, depression, and social isolation.[4,5] Overall, people with ET can have a decline in quality of life because of its impact on functional abilities and psychosocial considerations.[6] The goal of treatment for ET is to minimize functional disability, and thus should be customized based on the patient's individual needs and tolerance.

The International Parkinson and Movement Disorder Society (MDS) commissioned an Evidence-Based Medicine (EBM) subcommittee to assess the available evidence on therapeutic options in ET. The first MDS-sponsored EBM review for ET appraised published evidence of commercially available treatments until December 2016.[7] One of the conclusions of the review was the need to improve study design in ET trials to overcome the limitation of small sample sizes, crossover studies, short-term follow-up studies, and use of nonvalidated clinical scales.[7]

In 2021, the MDS EBM Committee developed a modified version of the Grading of Recommendations, Assessment, Development, and Evaluation (GRADE) framework to support future MDS EBM reviews on evidence interpretation and on drawing conclusions (implications for clinical practice) incorporating the uncertainty in the evidence and the magnitude of the clinical benefit.[8,9] More recent published MDS EBM reviews have already incorporated this methodology.[10,11]

In this review, we appraised the evidence available for interventions for ET with a focus on longer treatment effects. We started by updating the results from the prior MDS EBM review project from 2016 up to 2024, applying the revised appraisal scheme and increased benchmark to a minimum of 1 month of follow-up. In this review, we included only medical treatments, and a subsequent review on surgical treatments is under preparation.

Materials and Methods

We defined the research question using the PICOT approach, where population (P) included individuals with a clinical diagnosis of ET as reported by authors, interventions (I) encompassed any commercially available pharmacological therapy, comparators (C) were placebo or active comparator arms for pharmacological interventions. Outcomes (O) of interest were the severity, disability of tremors, or impairment in activities of daily living or functioning and quality of life related to tremor using clinical rating scales. Importantly, we defined time (T) as a minimum of 4 weeks of follow-up.

Search Strategy

To ensure a thorough and systematic method for identifying relevant studies, we developed the search strategy with the support of a library technologist and included the use of targeted keywords and databases to find studies that aligned with the eligibility criteria to be included in the review (see later inclusion and exclusion criteria). The detailed search strategy is presented in Supporting Information Appendix S1. Literature searches were done using electronic databases, including MEDLINE (1946 to April 2024), the CENTRAL database in the Cochrane Library (1946 to April 2024), and systematic checking of reference lists published in review articles and included studies (Fig. S1).

Selection of Studies

The search results were imported into COVIDENCE,[12] and screening of abstracts was done independently by two members of the subcommittee (T.A.M., D.D.) following predefined eligibility criteria. Conflicts were resolved by discussion. If consensus was not reached, the article was included in the next phase of the selection process. Inclusion criteria for studies in the review were: (1) randomized controlled trial (RCT) regardless of a parallel or crossover design; (2) individuals with a clinical diagnosis of ET; (3) a pharmacological agent, commercially available in at least one country; (4) intervention duration of at least 4 weeks; and (5) reporting of at least one of the following types of clinical outcome: tremor severity, functional impairment or disability, and/or health-related quality of life using a clinical rating scale developed for tremor. Exclusion criteria for studies were: (1) nonrandomized and/or noncontrolled study design; (2) an ET diagnosis not stated or unclear; (3) use of nonclinical outcome measures (eg, accelerometric data); (4) length of follow-up not specified; and (5) abstracts, reviews, or book chapters.

Data Extraction

Pairs of members of the subcommittee independently conducted data extraction from included studies, using a structured data collection form. Abstracted data included study design, participants' core eligibility criteria, primary and secondary outcomes, a description of the study and control interventions, daily dose, the number of participants allocated to the study and control interventions, duration of intervention, description of outcome measure used in the study, and the time point for outcome assessment. The studies reported outcomes in various ways, including mean or median differences between groups, changes from baseline, or the number of patients experiencing specific events in each

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group. These outcomes were captured along with details such as mean values, median values, confidence intervals (CIs), and *P* values.

Risk of Bias Assessment

We used the Cochrane Risk of Bias tool[13] that included the following domains: selection bias (such as randomization, allocation concealment, and random differences between groups as a result of small sample size), performance bias (blinding of participants and study personnel), attrition bias (incomplete outcome data), detection bias (modifications to validated rating scales and blinding of outcome assessment), reporting bias (selective reporting of results), and other biases. For interventions represented by only a single study, inability to assess inconsistency was considered and evidence was downgraded accordingly. For crossover RCTs, we considered the additional risk of bias derived from period and carryover effects.[14] For each study, two committee members rated risk of bias as high, low, or unclear for each domain. Any disagreements were resolved through consensus, or if necessary, by involving the entire subcommittee for a consensus. The overall risk of bias was rated as high if at least one domain was judged to have a "high risk of bias," or if there were multiple domains with "unclear risk" that altogether reduced the confidence on the reported results.[14]

Evidence Conclusions

The MDS-EBM program developed a framework to allow for efficacy and safety conclusions to be drawn from the balance between the GRADE level of evidence that considers risk of bias, imprecision, inconsistency, indirectness and likelihood of publication bias, and the importance of the benefit/harm (functional impairment) or lack thereof based on the clinical relevance of the outcomes. In this review, we primarily evaluated (and report on) efficacy. The efficacy assessments were primarily based on clinical and statistical significance derived from null hypothesis testing (*P* values), rather than effect size estimate. We calculated CIs from reported variances, where available.

Results

The literature search yielded a total of 1149 articles. After title and abstract selection, 114 articles remained and, after full-text article screening, 32 studies were included evaluating 20 different pharmacological interventions, of which 27 reports and 16 interventions included the use of placebo as comparator. The flowchart of the search and screening procedures is shown in Figure S1. Figure S2 illustrates the risk of bias assessment for interventions evaluated in placebo-controlled RCTs. Table 1 summarizes the efficacy conclusions (implications for clinical practice) for the 16 interventions evaluated against placebo and the corresponding evidence appraisal scheme. Efficacy conclusions were primarily driven by the results findings on tremor severity as the outcome domain more frequently reported across the included studies. We flagged safety concerns if contributing to the critical appraisal. Description of each study for agents that were evaluated in placebo-controlled RCT has been tabulated in Table S1.

Propranolol

Efficacy Conclusion

The reported results documented a statistically significant benefit of propranolol over placebo in reducing tremor severity. The overall certainty in the evidence was rated as low because of a high risk of bias and serious concerns about imprecision, leading to the conclusion of insufficient evidence to support the efficacy of propranolol in reducing tremor severity in ET.

Summary

Four placebo-controlled randomized trials[15-18] (parallel design, n = 1) assessed the efficacy of propranolol in reducing tremor severity after 4–8 weeks, with daily doses ranging from 120 to 240 mg. In total, 71 participants received propranolol and 72 received placebo (study sample size: 5–33). All studies reported statistically significant improvements in tremor severity with propranolol using clinician- and patient-rated ad hoc assessments used for the purpose of the study. Three studies evaluated the improvement in tremor severity by comparing scores of changes from baseline to end of treatment.[15,16,18] One study reported results as mean difference.[17] The overall risk of bias was high due to an unclear risk of bias in domains such as allocation concealment, selective reporting, and lack of information about wash-over period and period effect for crossover studies. Three of the included studies did not report CIs or detailed measures of variability, which together with small sample sizes led to serious concerns about imprecision. Publication dates of included studies ranged from 1973 to 1983.

Primidone

Efficacy Conclusion

The reported results documented a statistically significant benefit of primidone over placebo in reducing tremor severity. The overall certainty in the evidence was rated as low because of a high risk of bias and serious concerns about imprecision, leading to the conclusion that there is insufficient evidence to support the efficacy of primidone in reducing tremor severity in ET.

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Summary

Three placebo-controlled crossover randomized trials[19–21] assessed the efficacy of primidone in reducing tremor severity up to 5 weeks, with daily doses ranging from 125 to 750 mg, in a total of 60 participants (study sample size: 16–22). The studies reported statistically significant improvements in tremor severity with primidone using clinician- and patient-rated ad hoc assessments. The included studies had an overall high risk of bias because of unclear risk of bias in many domains, including allocation concealment, selective reporting, and incomplete outcome data. All three crossover studies lacked information about period effect. Studies were small, and two of the studies did not report measures of variability or CIs, leading to serious concerns about imprecision. The included studies were published in 1985,[19] 1988,[20] and 2000.[21] Notably, thirteen participants dropped out from the primidone arm because of acute toxic reaction, and no post-baseline efficacy data under primidone were available for them,[19] raising concerns for incomplete outcome data. In addition, a total of 22 participants taking primidone reported other side effects, including sedation, tiredness, nausea, vertigo, and depression, compared with participants in the placebo arm.

Topiramate

Efficacy Conclusion

The reported results documented a statistically significant benefit of topiramate over placebo in reducing tremor severity. The overall certainty in the evidence was rated as low because of a high risk of bias and serious concerns about imprecision, and therefore lead to the conclusion that there is insufficient evidence to support the efficacy of topiramate in reducing tremor severity in ET.

Summary

Two reports[22,23] in a total of four placebo-controlled randomized trials (parallel design, n = 1[22]) assessed the efficacy of topiramate in reducing tremor severity using the Fahn–Tolosa–Marin (FTM)[24] tremor rating scale after 10 or 24 weeks, with a mean daily dose of 215 or 296 mg. A total of 172 participants were allocated to the topiramate group and 156 to the placebo group (study sample size: 66 [pooled sample of three studies][23] and 223[22]). The available reports indicate statistically significant improvements in overall FTM tremor rating scale, but not consistently in tremor severity with topiramate. Both reports had an overall high risk of bias because of incomplete outcome data with higher rates of withdrawals in the topiramate arm (range: 37.6%[22] to 29%[23]) in comparison with the placebo arm (range: 20.9%[22] to 16%[23]), mostly related to common adverse effects such as cognitive difficulties, dizziness, and sedation. There were serious concerns about imprecision considering the study sample size,[23] and the reported CI included a negligible clinical benefit, because an overall FTM tremor score change of <5 points is considered of "no clinical improvement."[22] The included studies were published in 2006[22] and 2008.[23]

Alprazolam

Efficacy Conclusion

The reported results documented a statistically significant benefit of alprazolam over placebo in reducing tremor severity. The overall certainty in the evidence was rated as very low because of the high risk of bias, serious concerns about imprecision, and inconsistency could not be assessed. Therefore, we conclude that there is insufficient evidence to support the efficacy of alprazolam in reducing tremor severity in ET.

Summary

One placebo-controlled randomized crossover trial[21] assessed the efficacy of alprazolam (mean dose: 0.75 mg/day) in reducing tremor severity using exclusively an ad hoc tremor scale based on functional items, with a total of 22 participants receiving alprazolam for 4 weeks. There was a significant improvement in functional aspects of tremor with alprazolam. This study was judged to be at a high risk of bias because of unclear risk on many domains, including incomplete outcome data, selective reporting, and lack of information on period and carry-over effect.

Botulinum Toxin Type A

Efficacy Conclusion

The reported results documented a statistically significant benefit of botulinum toxin type A (BtA) over placebo in reducing tremor severity. The overall certainty in the evidence was rated as very low because of a high risk of bias and serious concerns about imprecision and inconsistency. In conclusion, there is insufficient evidence to support the efficacy of BtA injections in reducing tremor severity in ET.

Summary

Four placebo-controlled randomized trials[25–28] (parallel design, n = 3) assessed the efficacy of BtA injection (incobotulinumtoxinA or onabotulinumtoxinA) in reducing tremor severity up to 24 weeks with variable regimens of fixed and flexible dosing (mean study dose ranging between 50 and 116 I.U.).[25–28] In total, 221 participants were included, of which 140 received BtA injections (study sample size: 25–88). Studies reported improvements (albeit not consistently) in tremor severity with BtA compared with placebo, using various

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TABLE 1 *Summary of findings and efficacy conclusions of included interventions in the MDS-EBM review on medical treatments for essential tremor*

Intervention	RCT (n)	Treated (n)	Tremor severity (versus placebo)	Risk of bias	Precision	Consistency	Directness	Level of the evidence in favor of or against the intervention	Efficacy conclusion
Propranolol	4	71	Improved	High	Imprecise	Consistent	Direct	Low	Insufficient evidence
Primidone	3	60	Improved	High	Imprecise	Consistent	Direct	Low	Insufficient evidence
Topiramate	4	172	Improved	High	Imprecise	Consistent	Direct	Low	Insufficient evidence
Alprazolam	1	22	Improved (functional)	High	Imprecise	Unknown	Direct	Very low	Insufficient evidence
Botulinum toxin type A	4	140	Improved	High	Imprecise	Inconsistent	Direct	Very low	Insufficient evidence
Perampanel	1	22	Improved	High	Imprecise	Unknown	Direct	Very low	Insufficient evidence
Acetazolamide	1	22	Not improved	High	Imprecise	Unknown	Direct	Very low	Insufficient evidence
Flunarizine	1	17	Improved	Low	Imprecise	Unknown	Direct	Low	Insufficient evidence
Gabapentin	1	25	Improved	High	Imprecise	Unknown	Direct	Very low	Insufficient evidence
Levetiracetam	2	27	Not improved	High	Imprecise	Consistent	Direct	Low	Insufficient evidence
Mirtazapine	1	16	Not improved	High	Imprecise	Unknown	Direct	Very low	Insufficient evidence
Phenobarbitone	3	45	Not improved	High	Imprecise	Inconsistent	Direct	Very low	Insufficient evidence
Pregabalin	3	46	Not improved	High	Imprecise	Consistent	Direct	Low	Insufficient evidence
Progabide	1	10	Not improved	High	Very imprecise	Unknown	Direct	Very low	Insufficient evidence
Trazodone	2	24	Not improved	High	Imprecise	Consistent	Direct	Low	Insufficient evidence
Zonisamide	1	10	Not improved	High	Imprecise	Unknown	Direct	Very low	Insufficient evidence

Abbreviations: MDS-EBM, International Parkinson and Movement Disorder Society Evidence-Based Medicine; RCT, randomized controlled trial.

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rating scales. The overall risk of bias was considered high because of the risk of unblinding (incidence of hand weakness as side effect); hand weakness was reported more frequently (up to 30%–50%) with higher doses of botulinum toxin.[25,26] There were serious concerns about inconsistency, and imprecision due to wide margins of CI available in three studies.[25,26,28]

New Commercially Available Interventions since the 2019 EBM-MDS ET Review

Perampanel

Efficacy Conclusion

The reported results documented a statistically significant benefit of perampanel over placebo in reducing tremor severity. The overall certainty in the evidence was rated as very low because of a high risk of bias and serious concerns about imprecision. Furthermore, inconsistency could not be assessed. Therefore, we conclude that there is insufficient evidence to support the efficacy of perampanel in reducing tremor severity in ET.

Summary

One placebo-controlled randomized crossover trial[29] assessed the efficacy of perampanel in reducing tremor severity after 14 weeks, using a video-rated version of the Essential Tremor Rating Assessment Scale–Performance Subscale (TETRAS-P).[29] A total of 22 study participants were administered perampanel. There was an improvement in tremor severity in the video-rated TETRAS-P and in subject impression of change at a mean daily dose of 6 mg.[29] There was a high risk of bias overall due to a high risk for incomplete outcome data (36% missing in perampanel vs. 10% in placebo arm).[29] The period effect was significant for Quality of Life in Essential Tremor Questionnaire (QUEST) score, but not for other outcomes (uncertain risk). The sample size was small and the calculated CI was wide, leading to a serious concern for imprecision.

Other Interventions

Acetazolamide

Efficacy Conclusion

The reported results did not document a statistically significant benefit of acetazolamide over placebo in reducing tremor severity. The overall quality of evidence was rated as very low because of a high risk of bias and serious concerns about imprecision. Furthermore, inconsistency could not be assessed. There is insufficient evidence to support the efficacy of acetazolamide in reducing tremor severity in ET.

Summary

One placebo-controlled randomized crossover trial[21] assessed the efficacy of acetazolamide (mean daily dose: 562 mg) in reducing tremor severity using exclusively an ad hoc scale based on functional items. A total of 22 participants received acetazolamide for 4 weeks. There was no significant difference in tremor severity with acetazolamide compared with placebo. The overall risk of bias was high due to uncertain risk on many domains, including incomplete outcome data and lack of information on carry-over and period effects, and selective reporting. Due to the small sample size, there were serious concerns about imprecision. Acetazolamide was associated with paresthesia in 10% of participants.

Flunarizine

Efficacy Conclusion

The reported results documented a statistically significant benefit of flunarizine over placebo in reducing tremor severity. The certainty in the evidence was rated as low because of serious concerns about imprecision. Inconsistency could not be assessed. There is insufficient evidence to support the efficacy of flunarizine in reducing tremor severity in ET.

Summary

One placebo-controlled randomized crossover trial[30] assessed the efficacy of flunarizine in reducing tremor severity after 4 weeks, using an ad hoc rating scale. A total of 17 participants received flunarizine. There was an improvement in tremor severity, and 13 of 15 participants reported an improvement in tremor while taking flunarizine. The overall risk of bias was considered low. The study sample was small, and there were no reported measures of variance, thus raising serious concerns about imprecision. Two participants experienced mild sedation on flunarizine.

Gabapentin

Efficacy Conclusion

The reported results documented a statistically significant benefit of gabapentin over placebo in reducing tremor severity. The overall certainty in the evidence was rated as very low because of a high risk of bias and serious concerns about imprecision. Furthermore, inconsistency could not be assessed. There is insufficient evidence to support the efficacy of gabapentin in reducing tremor severity in ET.

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Summary

One placebo-controlled randomized crossover trial[31] assessed the efficacy of gabapentin in reducing tremor severity after 6 weeks, with two daily doses of 1800 and 3600 mg. A total of 25 participants participated in the study. The study reported improvements in observed tremor severity with gabapentin (regardless of the dose) using the Unified Tremor Rating and Assessment scale[24] and in the patient-reported activities of daily living section of the same scale. The overall risk of bias was high because of uncertain risk on domains of incomplete outcome data and selective reporting. A sample size of 25 and a wide calculated CI (mean difference of 2.7; CI = 0.49–4.91) raised serious concerns about imprecision. Adverse events observed in the gabapentin period included lethargy/drowsiness, fatigue, decreased libido, dizziness, nervousness, and shortness of breath.

Levetiracetam

Efficacy Conclusion

The reported results did not document a statistically significant benefit of levetiracetam over placebo in reducing tremor severity. The overall quality of evidence was rated as low because of a high risk of bias and serious concerns about imprecision. There is insufficient evidence to support the efficacy of levetiracetam in reducing tremor severity in ET.

Summary

Two placebo-controlled randomized crossover trials[32,33] enrolling a total of 27 participants assessed the efficacy of levetiracetam (mean daily dose: 3000 mg) in reducing tremor severity after 6 or 9 weeks, using the Location-Severity subscale of Tremor Rating Scale[24] and its total score. Both studies reported no improvement in tremor severity with levetiracetam. Overall risk of bias was high due to the domain incomplete outcome data. Adverse effects reported included fatigue, drowsiness, impaired balance, depressed mood, and dizziness.

Mirtazapine

Efficacy Conclusion

The reported results did not document a statistically significant benefit of mirtazapine over placebo in reducing tremor severity. The overall certainty in the evidence was rated as very low because of a high risk of bias and serious concerns about imprecision. Furthermore, inconsistency could not be assessed. There is insufficient evidence to support efficacy of mirtazapine in reducing tremor severity in ET.

Summary

One placebo-controlled randomized crossover trial[34] assessed the efficacy of mirtazapine (mean daily dose: 45 mg) in reducing tremor severity after 4 weeks, using the FTM rating scale.[24] A total of 16 participants received mirtazapine. There was no significant improvement in tremor severity with mirtazapine. The study's overall risk of bias was considered to be high because of incomplete outcome data and high risk about carryover effect.

Phenobarbitone

Efficacy Conclusion

The reported results did not document a statistically significant benefit of phenobarbitone over placebo in reducing tremor severity. The overall certainty in the evidence was very low due to an overall high risk of bias and serious concerns about imprecision and inconsistency. There is insufficient evidence to support the efficacy of phenobarbitone in reducing tremor severity in ET.

Summary

Three placebo-controlled randomized trials[18,20,35] (crossover design, n = 3) evaluated the efficacy of phenobarbitone in reducing tremor severity after 4–5 weeks, using ad hoc rating scales in a total of 45 participants. The daily dose of phenobarbitone ranged from 125 to 136 mg or 1.25 + 0.10 mg/kg. One study[35] reported significant improvement in tremor severity on a clinical rating scale, but not in a self-assessment scale. Two studies[18,20] reported nonsignificant changes in tremor severity, leading to concerns about consistency across study findings. Overall, the risk of bias was high due to multiple domains with unclear risk including incomplete outcome data and crossover and period effects. Two of the studies did not report measures of variability or CIs, and the third study had wide calculated CIs leading to serious concerns about imprecision.

Pregabalin

Efficacy Conclusion

The reported results did not document a statistically significant benefit of pregabalin over placebo in reducing tremor severity. The overall certainty in the evidence was rated as low because of a high risk of bias and serious concerns about imprecision. There is insufficient evidence to support the efficacy of pregabalin in reducing tremor severity in ET.

Summary

Three placebo-controlled randomized trials (parallel design, n = 1)[36–38] assessed the efficacy of pregabalin up to 600 mg/day in reducing tremor severity up to

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6 weeks, using the FTM tremor rating scale. A total of 46 participants received pregabalin. There was no significant improvement in tremor severity with pregabalin. There were high dropout rates (>20%) across studies. The overall risk of bias was considered high because of incomplete outcome data in studies.

Progabide

Efficacy Conclusion

The reported results did not document a statistically significant benefit of progabide over placebo in reducing tremor severity. The overall certainty in the evidence was rated as very low because of a high risk of bias and very serious concerns about imprecision. Furthermore, inconsistency could not be assessed. There is insufficient evidence to support the efficacy of progabide in reducing tremor severity in ET.

Summary

One placebo-controlled randomized crossover trial[39] assessed the efficacy of progabide (30 mg/kg/day) after 6 weeks in reducing tremor severity using an ad hoc severity scale in a total of 10 participants. There was no significant improvement in tremor severity with progabide. The overall risk of bias was considered high because of uncertain risk with blinding and incomplete outcome data domains, as well as with period or crossover effect.

Trazodone

Efficacy Conclusion

The reported results did not document a statistically significant benefit of trazodone over placebo in reducing tremor severity. The overall certainty in the evidence was rated as low because of a high risk of bias and serious concerns about imprecision. There is insufficient evidence to support the efficacy of trazodone in reducing tremor severity in ET.

Summary

Two placebo-controlled randomized crossover trials assessed the efficacy of trazodone (mean daily dose: 150 mg) in reducing tremor severity after 4[40] or 6[41] weeks using ad hoc rating scales. A total of 24 participants were treated with trazodone. Both studies reported no significant improvements in tremor severity with trazodone compared with placebo. The overall risk of bias was high due to high risk in incomplete outcome data, and uncertain risk in carryover effect and selective reporting. The studies were small and did not report on measures of variance, leading to serious concerns about imprecision.

Zonisamide

Efficacy Conclusion

The reported results did not document a statistically significant benefit of zonisamide over placebo in reducing tremor severity. The overall certainty in the evidence was rated as very low because of a high risk of bias and serious concerns about imprecision. Furthermore, inconsistency could not be assessed. There is insufficient evidence to support the efficacy of zonisamide in reducing tremor severity in ET.

Summary

One placebo-controlled randomized parallel trial[42] assessed the efficacy zonisamide (mean dose: 160 mg/day) in reducing tremor severity after 4 weeks, using the FTM tremor rating scale. A total of 10 participants received zonisamide. There was no significant improvement in tremor severity with zonisamide. The risk of bias was high overall due to incomplete outcome data. Zonisamide was associated with a 30% dropout rate as a result of adverse events.[42] The sample size was small, and the calculated CI was very wide, raising serious concerns about imprecision.

Discussion

In this updated EBM review of pharmacological treatments for ET, we found reports of improvement of tremors compared with placebo after a minimum of 1 month of follow-up with propranolol, primidone, topiramate, and BtA in more than one study. Studies of flunarizine, gabapentin, and perampanel reported positive findings but only in a single study. Following the GRADE framework for evidence interpretation and drawing conclusions for clinical practice adapted by MDS, we concluded that there was insufficient evidence to confidently reach a (likely) efficacious recommendation for any of these interventions, including those more commonly used in clinical practice, such as propranolol, primidone, or topiramate.

It is worth noting that in the 2019 MDS EBM ET review, propranolol, primidone, and topiramate were considered to be "clinically useful" for treating people with ET. Without novel trials included in this review for these interventions, the differences in efficacy conclusions between the current update and the first-ever MDS EBM review on ET treatment can be explained in light of various considerations. First, we have now defined as inclusion criterion a minimum duration of 4 weeks for a treatment period, which can be considered an advance from the prior review for what is the ideal duration of a clinical trial evaluating a symptomatic intervention for a chronic movement disorder such as ET. This approach led to the exclusion of previously included studies and

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significantly reduced the data available for appraisal, which may have lowered the confidence in the available evidence. We believe that a threshold of 4 weeks for trial duration is nonetheless conservative, because longer treatment periods of 12 weeks have been required for chronic movement disorders, such as PD, in other MDS-EBM reviews.[9,10] An immediate implication of this finding is the urgent need for longer and better-designed clinical trials in ET, a trend that has fortunately begun to emerge in more recent studies.

Second, the GRADE approach adopted and adapted by the MDS EBM program is now widely recognized as the standard to support conclusions based on the evidence available in our field. This framework involves a rigorous, standardized, and transparent assessment of the quality of the body of evidence and the clinical relevance of the findings. In this review, we concluded that, for all interventions, the certainty in the evidence was low to very low and, therefore, the evidence is considered insufficient to reach a conclusion of "not efficacious," "efficacious," or "likely efficacious." Most included studies in this review predate the establishment of reporting guidelines such as the Consolidated Standards of Reporting Trials, introduced in 1996[43] to improve reporting of RCTs, and indirectly the standards for study design and conduction. It is therefore not surprising that reports of the included trials lacked sufficient details about randomization, allocation concealment, and incomplete/nonexistent description of study flow. Consequently, assessment of the risk of bias was frequently challenging and commonly led to a conclusion of high risk. The concerns about imprecision were more commonly motivated by many interventions being evaluated in small underpowered studies, and CIs not being accessible (Table S2). In other words, the conclusion of "insufficient evidence" for all interventions was a direct consequence of the quality of the studies available in ET included in this update. We would like to emphasize that GRADE does not "reject" therapeutic interventions but highlights strengths and limitations in evidence quality to confidently support a claim of therapeutic effectiveness.

This review does not include other interventions, such as atenolol,[44-46] which is frequently prescribed for ET[47-49] despite the absence of robust controlled trial data, thus emphasizing the need for rigorous clinical trials to assess their efficacy.

This review is not intended to serve as a guideline for the treatment of ET but rather to provide an evidence-based synthesis of the available interventions and their efficacy using the best quality available data and a rigorous evaluation of the strengths and limitations. The current conclusions should be considered one of many sources of information prescribing healthcare professionals may consider for management decisions an in ET. Along these lines, the 2019 ET MDS EBM review remains useful to a certain extent, being another source of information that includes the evidence available for a shorter treatment duration.

More importantly, the current results are an important call for future research on ET therapeutics to follow contemporary standards in clinical trials methodology with adequately powered studies that use clinically meaningful outcome measures and have adequate follow-up durations to provide robust information on therapeutic efficacy and safety for a chronic condition such as ET. In Table S2, we have listed these and other best practices applicable to various aspects of study design in future ET trials.

In the context of integrating EBM results in clinical practice, it is important to consider patient values, treatment costs, and feasibility into therapeutic decision-making. Although the current results indicate low to very low certainty for included interventions, "classical" options, such as propranolol and primidone, remain relatively low-cost and widely available. Moreover, patient preferences vary substantially with respect to adverse effect profiles, dosing complexity, and perceived symptom relief, meaning that the individual acceptability of an intervention may differ despite similar evidence ratings. Taken together, the present evidence summaries should be used in conjunction with other sources of information for shared decision-making, rather than serving as the sole determinant of care choices. ■

Author Roles: 1. Research project: A. Conception, B. Organization, C. Execution; 2. Statistical Analysis: A. Design, B. Execution, C. Review and Critique; 3. Manuscript: A. Writing of the first draft, B. Review and Critique.
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Acknowledgments: We are thankful to Shelby Shuster from the MDS secretariat for the administrative support to this project.

Financial Disclosures and Conflicts of Interest: Deepa Dash has received honorarium for speaking from Merz Therapeutics and a grant from the Parkinson's Society of Canada and Southwestern Parkinson's Society. Verónica Bruno has received research support from Parkinson Canada, Parkinson Foundation, Calgary Health Foundation, and the Hotchkiss Brain Institute. Petra Schwingenschuh has received consulting and advisory board membership with honoraria from AbbVie, Bial, Merz, and Stada. Kelly E. Lyons has no disclosures. Eng-King Tan has received honorarium from Elsevier for editorial duties. Claudia M. Testa has received honoraria for clinical trial steering committee work sponsored by Neurocrine and uniQure, and speaker honorarium from the Virginia Neurological Society; scientific consulting funds from LifeEdit, LoQus23, and Klick Health; book coeditor royalties from Oxford University Press; research funding from the Parkinson's Foundation (PD GENEration), Sage Therapeutics, and AskBio (clinical trial contracts with UNC-CH); and

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salary from the University of North Carolina at Chapel Hill. Santiago Perez Lloret has received honorarium from Elea Laboratories and has received grants from the Agencia Científica y Técnica, Argentina, and the Pontificia Universidad Católica Argentina; owns stock options from TeleNeuro Solutions LLC; and has received salary from National Scientific and Technic Research Council of Argentina. Bettina Balint has no disclosures. João Costa is co-chair of the guidelines production group of the European Academy of Neurology. Rob M.A. de Bie has received research grants from Amsterdam Neuroscience, Amsterdam UMC Foundation, Medtronic, ROMO Foundation, Stichting ParkinsonFonds, Stichting Universitas, ZonMw, and Edmond J. Safra Fellowship from International Parkinson and Movement Disorder Society (MDS), all paid to the institution; and has received salary from Amsterdam UMC. Monty A. Silverdale has received grants from Parkinson's UK, The Michael J. Fox Foundation, and Medical Research Council; and has received honoraria for lecturing and consulting from Bial and Medtronic. Ai Huey Tan has received consultancy fees from Elsevier as Associate Editor for *Parkinsonism and Related Disorders*; speaker honoraria from Eisai, Orion Pharma, and MDS; and has received research support from The Michael J. Fox Foundation and the Global Parkinson Genetics Program. Tiago A. Mestre has received consulting and Advisory Board Membership with honoraria: AbbVie, MDS, AAN, CHDI Foundation/Management, Roche, UCB, PTC Therapeutics, Novartis, WAVE therapeutics; grants and research: EU Joint Program – Neurodegenerative Disease Research, uOBMRI, Roche, Ontario Research Fund, CIHR, The Michael J. Fox Foundation for Parkinson's Research, Parkinson Canada, Parkinson Research Consortium, and Brain Canada; and salary from UOMA.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Data

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